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### Measuring device for determining a distribution of a heat transfer medium and method for determining a distribution of a heat transfer medium

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#### Abstract

A measuring device for determining a distribution of a heat transfer medium on an inner wall of a shaftless container which rotates when used to heat the heat transfer medium with concentrated solar radiation in a solar thermal power plant or as a rotary kiln includes a distance measuring device for determining a thickness of a film of the heat transfer medium on the inner wall of the container. The distance measuring device includes at least one optical device for detecting at least one height profile along at least one measurement line projected onto the inner wall and at least one position transducer for determining a current rotational position of the respective measurement line on the inner wall. A method for determining a distribution of a heat transfer medium on an inner wall of a shaftless container is also provided.

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## Background/Summary

### BACKGROUND AND SUMMARY

(1) The invention relates to a measuring device for determining a distribution of a heat transfer medium in a receiver device for solar radiation in a solar thermal power plant and to a method for determining a distribution of a heat transfer medium.

(2) DE 102010062367 A1 describes a device for receiving solar radiation. The device comprises a container with an inner wall and a rotational drive device, by which the container is rotated about an axis of rotation. The container has an axis which is oriented parallel or at an acute angle to the direction of gravity. A heat transfer medium is guided through the container along the inner wall to form a heat transfer medium film. Particles or a liquid, for example, are proposed as the heat

transfer medium in DE102010062367 A1.

(3) In experiments with the device described, there were indications of an inhomogeneous distribution of the film and thus of the heat transfer medium on the inner wall of the device. A possible reason for this can be that the surface of the inner wall of the container can change during operation due to the high thermal load of solar radiation and thermal cycles between times with solar radiation and times without solar radiation, so that a homogeneous distribution of the film may no longer be given.

(4) In order to recognize this, a simulation of the film thickness of the heat transfer medium is usually carried out, which however involves many simplifications and assumptions that are not fully validated.

(5) It is desirable to create a measuring device for determining a distribution of a heat transfer medium in a receiver device for solar radiation in a solar thermal power plant, which provides meaningful data.

(6) It is also desirable to specify methods for determining a distribution of a heat transfer medium using such a measuring device.

(7) A measuring device is proposed for determining the distribution of a heat transfer medium on an inner wall of a shaftless container that rotates when used as intended, which is provided in particular to heat the heat transfer medium with concentrated solar radiation in a solar thermal power plant or as a rotational kiln. The measuring device includes a distance measuring device for determining a thickness of a film of the heat transfer medium on the inner wall of the container. The distance measuring device comprises at least one optical device for detecting at least one height profile along at least one measurement line projected onto the inner wall and at least one position transducer for determining a current rotational position of the respective measurement line on the inner wall relative to a rotational position of the container.

(8) The container is in particular a so-called particle receiver in a solar thermal power plant. In a particle receiver, solar rays are concentrated in the interior and a particulate heat transfer medium is heated, the particles of which are distributed along the inner wall of the particle receiver when the particle receiver is rotated. The particles form a particle film essentially parallel to the inner wall of the particle receiver. The particle film moves from an inlet into the particle receiver to an outlet from the particle receiver.

(9) The film thickness can advantageously be measured. A possibly incorrect calculation or simulation of the film thickness can be omitted. The film thickness can not only be assessed qualitatively, whether there are gaps in the film, for example, but a reliable statement can be made about the film thickness on the inner wall of the container in the measured region.

(10) The measuring device is particularly suitable for so-called centrifugal receivers, also known as particle receivers, in receiver devices for solar radiation in solar thermal power plants and for rotational kilns.

(11) According to an advantageous configuration of the measuring device, the distance measuring device can be designed to detect distance data directly on the inner wall of the container along the measurement line in order to create the height profile.

(12) A statement about the height profile on the entire surface of the inner wall or on regions of the inner wall that are of interest can be obtained if, for example, a large number of measurement lines lying next to one another are detected one after the other. The container can expediently be set in rotation for this purpose, so that a region or the entire surface of the inner wall can be measured.

(13) According to an advantageous configuration of the measuring device, the distance measuring device can be designed to detect distance data directly on the film of the heat transfer medium on the inner wall of the container along the measurement line in order to create the height profile.

(14) A statement about the height profile on the entire surface of the inner wall or on regions of the inner wall that are of interest can be obtained if, for example, a large number of measurement lines lying next to one another are detected one after the other. The container can expediently be rotated for

this purpose, so that a region or the entire surface of the inner wall can be measured.

(15) A light section can advantageously be used to detect the height profile. The light section is a known method of optical 3D measurement technology, with which a height profile can be measured along a projected line of light and is based on the principle of triangulation. The optical device can in particular be a light section sensor, which includes a line projector, usually with a laser as the light source, which projects a line that is as narrow and bright as possible onto the measurement object, and an electronic camera, which detects the projection of the measurement line on the object, in this case the clean inner wall or the film on the inner wall. The displacement of the measurement line in the camera image can be converted into 3-D coordinates using the methods of photogrammetry that are known per se. Measuring devices that use this method are known, for example, as so-called profile scanners.

(16) Profile scanners measure a height profile along a projected line of light, which represents the measurement line. The distance is not measured continuously. There is a certain number of measuring points along the height profile, each of which generates its own measured value for the distance.

(17) In the case of film thickness measurement in the solar radiation receiver, a profile scanner can be held in the container by means of a holder and a height profile of the particle film can be measured along the longitudinal axis of the container, also known as the receiver. This height profile is initially only a 2D profile. If the container now also rotates and the measurement of the previously measured height profile is continuously repeated, a 3D height profile of the film of the heat transfer medium is created.

(18) In the case of a perfect cylinder, the thickness of the particle film could be calculated from the 3D height profile and the determined distance of the profile scanner to the inner wall of the receiver. However, since the container is not a perfect cylinder, the measurement of the 3D profile alone does not provide any information about the actual thickness of the particle film.

(19) In order to determine this thickness, a reference measurement of the empty container without heat transfer medium can also be carried out. The inner wall of the container, also referred to as the inliner, is measured in the same way as the film of the heat transfer medium on the inner wall and a 3D height profile of the inner wall is created. The two height profiles can then be subtracted from one another. The result is the actual thickness of the film of the heat transfer medium.

(20) In order to be able to correctly subtract the two height profiles from one another, information about the rotational position of the two height profiles or the respective measurement line in relation to this rotational position of the container is also required. The rotational position of the measurement line correlates closely with the rotational position of the container, in particular the rotational position of the measurement line can be equal to the rotational position of the container. This allows the measurement points of both height profiles to refer to the same real point in the container.

(21) This information can be provided by the position transducer, also known as an encoder or encoder. This is a rotation angle sensor that can be used to determine the rotational position of the container.

(22) This rotational position can be transferred to the profile scanner so that the individual measuring points along the respective measurement line can be saved with the information about the associated rotational position of the container. Alternatively or additionally, the rotational position of the container can be transferred to a computing device, which can accordingly assign the measurement points to the information about the rotational position and the height profile of the associated measurement line.

(23) A transfer to the profile scanner can take place via an interface, for example via an RS-422 interface, which can be connected directly from the position transducer to the profile scanner. The measurement data on the rotational position can then be stored at the same time as the measurement data regarding the distance and can be transmitted to the computing device and processed.

(24) The heat transfer medium can advantageously consist of or comprise particles, for example bauxite particles. The height profile is measured on a surface of a film of the heat transfer medium in the form of a particle bed. It turns out that even a porous bed with the absorptive material of the bauxite particles reflects enough light.

(25) According to an advantageous embodiment of the measuring device, the measurement line can extend parallel to a longitudinal axis of the container. The measurement line consists of or comprises a number of individual measurement points along the length of the measurement line, in which the optical device determines the distance between the measurement object and the sensor for each measurement point, for example via a transit time measurement.

(26) The profile scanner can have a variable measuring range transverse to the longitudinal axis in the direction of the surface normal of the inner wall of the container. The measuring range can be adjusted depending on how different the distances to be measured are, for example due to non circular running or eccentricity of the container. With small changes in distance, the measuring range can be very small. If there are larger differences, the range can be chosen larger, for example if the circularity is not perfect or if there is an eccentricity.

(27) The measuring frequency of the profile scanner depends on the size of the measuring range. The smaller this range, the fewer measuring points have to be read out on the sensor and the faster measurements can be taken. Accordingly, it is desirable to keep this range small.

(28) An eccentrically running, rotating component can be problematic, since the measuring range would actually have to be quite large in order to be able to detect everything precisely. This can be solved using software technology by evaluating the data from the position transducer, which can be connected to the profile scanner and/or to the computing unit, together with the data of the height profiles of the profile scanner.

(29) In this case, a reference measurement can first be carried out in order to detect deviations in the circularity of the container. The information from the reference measurement can then be used to shift the measuring range according to the imperfect circularity of the container in the height profile. This has the advantage that the measuring range can be selected to be quite small, which results in a higher measuring frequency. Nevertheless, the distance information can be recognized with sufficient accuracy, since the measuring range is shifted according to the lack of roundness

(30) According to an advantageous embodiment of the measuring device, the position transducer can be coupled to the container, in particular attached to the container, and detect a rotational position of the container. In particular, the profile scanner can detect a rotational position of the container, which is supplied by the position transducer, synchronously with the projected measurement line. Alternatively or additionally, the position transducer can forward the rotational position to a computing unit, which combines the projected measurement line and the respective rotational position. The position transducer can advantageously have a magnetic tape that is placed around the container. In particular, the rotational position of the container can be determined using a magnetic tape with a number of magnetic poles, which are read out by a sensor of the position transducer that moves over the magnetic tape.

(31) A container, in particular a particle receiver, for heating the heat transfer medium of a receiving device for solar radiation or a rotational kiln usually does not have a shaft around which the container can rotate. Rather, it is driven by a chain. Therefore, conventional position transducers cannot be used to determine the rotational position of the particle receiver.

Advantageously, the rotational position can be determined by means of a magnetic tape placed around the container with a specific number of poles, which can be read by a sensor. This combination of sensor and magnetic tape can then be used to determine the rotational position of a shaftless container, in particular a particle receiver, with great accuracy.

(32) It is particularly advantageous to use a combination of a magnetic tape placed around the container and a sensor arranged above it for detecting a rotational position of a shaftless rotating particle receiver in a solar thermal power plant or in a rotational kiln. The sensor may be placed at a

fixed angular position with respect to the container. While the container rotates, the sensor does not rotate. The distance between the sensor and the magnetic tape is suitably between about 1 mm and about 3 mm. Optionally, the sensor can be mounted so that it can be moved radially in order to ensure an even distance to the receiver or the magnetic tape. This is advantageous if, for example, the container is not running circularly or is eccentric, or if the distances between the magnetic tape and the sensor can fluctuate.

(33) According to an advantageous embodiment of the measuring device, a computing device can be coupled to the distance measuring device and/or the position transducer.

(34) Advantageously, the computing device can carry out a corresponding assignment of the measuring points to the information about the rotational position of the container and the distance according to the height profile of the associated measurement line. In particular, in this case, height profile data from a measurement of the inner wall of the empty container without a heat transfer medium and from a measurement with a film of the heat transfer medium can be processed. The two height profiles can then be subtracted from one another. The result is the actual thickness of the film of the heat transfer medium.

(35) According to an advantageous configuration of the measuring device, the distance measuring device can protrude into the container on a holder. In particular, a plurality of the optical devices of the distance measuring device can protrude into the container on a holder. If the distance measuring device has only one optical device, in particular in the form of a profile scanner, this can be arranged to be displaceable along and/or with the holder so that the container can be measured over its entire length.

(36) If the distance-measuring device has a plurality of optical devices, in particular in the form of profile scanners, their number can expediently be selected such that the entire axial extent of the container can be detected with adjacent measurement lines.

(37) According to a further aspect of the invention, a method is proposed for determining a distribution of heat transfer medium on an inner wall of a shaftless container that rotates when used as intended, in particular in which the heat transfer medium of a receiver device for solar radiation in a solar thermal power plant or a rotational kiln is heated, by means of a measuring device, which comprises at least one optical device for detecting at least one height profile along at least one measurement line projected onto the inner wall and at least one position transducer for determining a current rotational position of the respective measurement line on the inner wall.

(38) The distance measuring device detects at least one height profile along at least one measurement line projected onto the inner wall using the at least one optical device, and the at least one position transducer determines a current rotational position of the container and thus the rotational position of the respective measurement line on the inner wall.

(39) According to an advantageous embodiment of the method, a rotational position of the two height profiles with and without a heat transfer medium on the inner wall of the container can be determined in relation to a rotational position of the container. As a result, the film thickness of the heat transfer medium on the inner wall of the container can be determined with precise positioning by subtracting the data.

(40) According to an advantageous embodiment of the method, the distance can be measured without solar radiation entering the container. Conventional components that do not tolerate high temperatures can advantageously be used.

(41) According to an advantageous embodiment of the method, a difference between the distance data of the film and the distance data of the inner wall can be formed and from this a location-dependent distribution of the film of the heat transfer medium on the inner wall of the container can be determined. As a result, reliable measured values can be provided.

(42) According to an advantageous embodiment of the method, a measuring frequency of the position transducer can be adapted to a length of the measurement line in the direction of the longitudinal axis of the container. As a result, an exact measurement can be carried out even in the

case of larger changes in distance, namely in the case of a profiled surface with a correspondingly pronounced height profile.

(43) According to an advantageous embodiment of the method, a reference measurement can be carried out to determine an eccentricity and/or lack of circularity of the container and a length of the measurement lines can be adjusted in the direction of the longitudinal axis of the container. The determination of the distribution of heat transfer medium on the inner wall of the container can advantageously be adapted to the actual state of the container.

(44) According to an advantageous embodiment of the method, the determination of the distribution of heat transfer medium on the inner wall can be carried out repeatedly and changes in the distribution can be detected. This allows a reliable quality check to be carried out during the service life of the solar radiation receiver.

(45) According to an advantageous embodiment of the method, a maintenance requirement can be indicated if permissible tolerances of the changes are exceeded. Timely maintenance or repair can favorably enable reliable operation of the solar radiation receiver.

(46) According to a further aspect of the invention, a use of a measuring device is proposed for determining a distribution of a heat transfer medium on an inner wall of a container which is designed to heat the heat transfer medium with concentrated solar radiation in a solar thermal power plant, comprising a distance measuring device for determining a thickness of a film of the heat transfer medium on the inner wall of the container, wherein the distance measuring device comprises at least one optical device for detecting at least one height profile along at least one measurement line projected onto the inner wall and at least one position transducer for determining a current rotational position of the respective measurement line on the inner wall, wherein the position transducer has a combination of a magnetic tape and a sensor arranged thereabove for detecting a rotational position of the shaftless rotating container.

(47) The container is in particular a particle receiver, also known as a centrifugal receiver, in a solar thermal power plant. Optionally, the combination can also be used to detect a rotational position of a rotational kiln.

(48) According to an advantageous embodiment, the magnetic tape can be arranged on the container and can rotate with the container, while the sensor is arranged above the magnetic tape. The magnetic tape can surround the container, in particular on the outer circumference, and the sensor can be arranged at a small radial distance from it. The sensor may be placed at a fixed angular position with respect to the container. While the container rotates, the sensor does not rotate. The distance between the sensor and the magnetic tape is suitably between about 1 mm and about 3 mm. Optionally, the sensor can be mounted so that it can be moved radially in order to ensure an even distance to the receiver or the magnetic tape. This is advantageous if, for example, the container is not running circularly or is eccentric, or if the distances between the magnetic tape and the sensor can fluctuate.

(49) Optionally, the magnetic tape can also be arranged on the inside, in particular on the inner wall, of the container and the sensor can be arranged correspondingly inside the container at a small radial distance from it.

(50) According to an advantageous embodiment, the magnetic tape can be arranged on the container and can rotate with the container, while the magnetic sensor is arranged above the container or magnetic tape. The sensor can be arranged on the outside of the container, in particular on the outer circumference, and the magnetic tape can be arranged at a small radial distance from it. Optionally, the magnetic tape can also be arranged on the inside, in particular on the inner wall, of the container and the sensor can be arranged correspondingly inside the container at a small radial distance from it.

(51) Optionally, the two alternative embodiments can also be combined on one container, wherein the measuring accuracy can be increased by two such combinations of sensor and magnetic tape.

(52) According to a further aspect of the invention, use of a combination of a magnetic tape and a

sensor arranged above it is proposed for detecting a rotational position of a shaftless rotating container. The container is in particular a particle receiver, also known as a centrifugal receiver, in a solar thermal power plant. Optionally, the combination can also be used to detect a rotational position of a rotational kiln.

(53) According to an advantageous embodiment, the magnetic tape can be arranged on the container and can rotate with the container, while the sensor is arranged above the magnetic tape. The sensor may be placed at a fixed angular position with respect to the container. While the container rotates, the sensor does not rotate. The distance between the sensor and the magnetic tape is suitably between about 1 mm and about 3 mm. Optionally, the sensor can be mounted so that it can be moved radially in order to ensure an even distance to the receiver or the magnetic tape. This is advantageous if, for example, the container is not running circularly or is eccentric, or if the distances between the magnetic tape and the sensor can fluctuate.

(54) The magnetic tape can surround the container, in particular on the outer circumference, and the sensor can be arranged at a small radial distance from it. Optionally, the magnetic tape can also be arranged on the inside, in particular on the inner wall, of the container and the sensor can be arranged correspondingly inside the container at a small radial distance from it.

(55) According to an advantageous embodiment, the sensor can be arranged on the container and can rotate with the container, while the magnetic sensor is arranged above the sensor. The sensor can be arranged on the outside of the container, in particular on the outer circumference, and the magnetic tape can be arranged at a small radial distance from it. Optionally, the magnetic tape can also be arranged on the inside, in particular on the inner wall, of the container and the magnetic tape can be arranged inside the container at a small radial distance from it.

(56) Optionally, the two alternative configurations can also be combined on one container, wherein the measuring accuracy can be increased by two such combinations of sensor and magnetic tape.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) Further advantages will be apparent from the following description of the drawings. Exemplary embodiments of the invention are shown in the figures. The figures, the description, and the claims contain numerous features in combination. A person skilled in the art will expediently also consider the features individually and combine them into further meaningful combinations.

(2) In the Exemplary Figures:

(3) FIG. 1 shows an exemplary embodiment of a container of a solar radiation receiver which is driven by a chain;

(4) FIG. 2 shows a schematic representation of a detection of a height profile on a measurement line;

(5) FIG. 3 shows a schematic representation of a detection of a height profile in a region of successive measurement lines in a circumferential direction according to FIG. 2;

(6) FIG. 4 shows a measuring device according to an exemplary embodiment of the invention, in which a distance measuring device has a displaceable optical device;

(7) FIG. 5 shows a measuring device according to an exemplary embodiment of the invention, in which a distance measuring device has a plurality of optical devices;

(8) FIG. 6 shows a perspective view of a container in a framework with a position transducer on a container according to an exemplary embodiment of the invention;

(9) FIG. 7 shows a detailed view of an arrangement of a magnetic tape of the position transducer according to FIG. 6 on a bearing flange of the container;

(10) FIG. 8 shows a detailed view of an arrangement of the position transducer with sensor and magnetic tape according to FIG. 6;



- (11) FIG. 9 shows a detailed view of a magnetic tape of a position transducer on a flange of a container according to an exemplary embodiment of the invention;
- (12) FIG. 10 shows a detailed view of the position transducer on a flange of a container according to FIG. 9;
- (13) FIG. 11 shows, in a cut-away perspective view, a container in a framework with an optical device on a holder according to an exemplary embodiment of the invention;
- (14) FIG. 12 shows a flow chart for carrying out a method according to an exemplary embodiment of the invention.

#### DETAILED DESCRIPTION

- (15) In the figures, identical or identically acting components are identified by the same reference signs. The figures only show examples and are not to be understood as limiting.
- (16) Before the invention is described in detail, it should be pointed out that it is not limited to the respective components of the device and the respective method steps, since these components and methods can vary.
- (17) The terms used herein are only intended to describe particular embodiments and are not used in a limiting manner. Furthermore, if the singular or indefinite articles are used in the description or in the claims, this also applies to the plural of these elements, unless the overall context clearly indicates otherwise.
- (18) The directional terminology used in the following with terms such as “left”, “right”, “above”, “below”, “in front of”, “behind”, “after”, and the like only serves for better comprehension of the figures and is in no way intended to restrict the generality. The components and elements shown, their design and use can vary according to the considerations of a person skilled in the art and can be adapted to the respective applications.
- (19) FIG. 1 shows an exemplary embodiment of a container **120** of a solar radiation receiver **100** which is driven by a chain (not shown).
- (20) The container **120** is double-walled and has an inner wall **130** on which a film **152** of a heat transfer medium **150**, for example bauxite particles, moves through the container **120** during intended operation. Along the axial extent **162** of the container **120**, the film **152** is strongly heated by the solar radiation **112** entering through the aperture **126** for radiation input. The heat transfer medium **150** is fed in, for example, via the medium inlet **124** on the opposite side of the container **120** in a manner known per se and is distributed by the rotation of the container **120** on its inner wall **130**.
- (21) The longitudinal axis **160** of the container **120** can be inclined at an angle **114** with respect to gravity *g*.
- (22) In order to determine the thickness of the film **152**, a distance measuring device **10** is used, which comprises at least one optical device **30** with a transmitter **32** and a receiver **34**. The optical device **30** is advantageously designed as a profile scanner, with which a measurement line **40** (FIG. 2) is directed onto the object to be measured, for example the inner wall **130** with or without film **152**.
- (23) FIG. 2 shows a schematic representation of the detection of a height profile by a distance measuring device **10** on a measurement line **40**, while FIG. 3 shows a schematic representation of the detection of a height profile on a region **42** of measurement lines **40** following one another in a circumferential direction **140** in accordance with FIG. 2.
- (24) The transmitter **32** is conveniently a laser and the receiver **34** is conveniently an electronic camera. The transmitter **32** of the optical device **30** radiates a laser beam onto the inner wall **150**, and the receiver **34** measures at points along a measurement line **40** the distances between the optical device **30** and points on the measurement line **40**.
- (25) During the measurement, the optical device **30** is rigidly mounted, while the container **120** can move in the direction of rotation **170** under the measurement line **40**. If the measurement line **40** or the region **42** of measurement lines **40** is positioned above a step in the film **152**, the height

difference **154** is detected, so that a height profile can be detected with the measurement line **40**.

(26) A height profile of the inner wall **130** without the film **152** and a height profile of the inner wall **130** with the film **152** of heat transfer medium **150** can be detected. From the difference in values, the thickness of the film **152** can be determined. A rotational position of the container **120** is determined for each measurement line **40**, so that the thickness of the film **152** can be determined with positional precision.

(27) FIG. **4** shows a measuring device **100** according to an exemplary embodiment of the invention, in which a distance measuring device **10** has an optical device **30** that can be displaced on or with a holder **20**.

(28) The distribution of heat transfer medium **150** on the inner wall **130** of the container **120** can be determined with the measuring device **100**. The measuring device **100** comprises the distance measuring device **10**, which in this exemplary embodiment comprises an optical device **30** for detecting at least one height profile along at least one measurement line **40** projected onto inner wall **130**, and at least one position transducer **50** for determining a current rotational position of the respective measurement line **40** on the inner wall **130** with respect to the rotational position of the container **120**.

(29) The measurement line **40** extends parallel to a longitudinal axis **160** of the container **120**. If the container **120** rotates, adjacent measurement lines **40** form a region **42**, as schematically shown in FIG. **3**.

(30) The position transducer **50** is attached to the container **120** and detects the rotational position of the container **120**, in particular synchronously with the projected measurement line **40** in a measuring position. The position transducer **50** can forward the position values to the distance measurement device **10** or, as indicated in the figure, to a computing device **90**, which links the distance measurement values and the position values.

(31) It goes without saying that the measured values can be transmitted from the position transducer **50** to the optical device **30** and/or to the computing unit **90**, as indicated by dot-dashed lines in FIGS. **4** and **5**, by wire via data cables or also wirelessly.

(32) Since the container **120** cannot be driven by a shaft the angular position of which could be detected when the container **120** rotates, the position transducer **50** has a magnetic tape **52** which is placed around the container **120**. In order to determine the rotational position, the magnetic tape **52** has a number of magnetic poles, which are read out by a sensor **54** of the position transducer **50**.

(33) In order to also detect the distribution of the heat transfer medium **150** on the entire inner wall **130** of the container **120** in the direction of the longitudinal axis **160**, the optical device **30** can be shifted parallel to the longitudinal axis of the container **120** to a new measurement position.

(34) FIG. **5** shows a measuring device **100** according to an exemplary embodiment of the invention, in which, in contrast to the exemplary embodiment in FIG. **4**, a distance measuring device **10** has a plurality of optical devices **30**, in this example three optical devices **30**. The number of optical devices **30** can be advantageously selected such that their measurement lines **40** projected onto the inner wall **130** or the film **152** adjoin one another in the direction of the longitudinal axis **160** of the container **120** and cover the entire axial extension **162** thereof. The optical devices **30** can then remain at their respective measuring point when determining the distribution of the heat transfer medium **150** on the inner wall **130**.

(35) FIG. **6** shows a perspective view of a container **120** in a framework **200** with a position transducer **50** on the container **120** according to an exemplary embodiment of the invention. FIG. **7** shows a detailed view of an arrangement of a magnetic tape **52** of the position transducer **50** according to FIG. **6** on a bearing flange **132** of the container **120**. FIG. **8** shows a detailed view of an arrangement of the position transducer **50** with sensor **54** and magnetic tape **52** according to FIG. **6**.

(36) The container **120** has no shaft for the rotation of the container **120** during normal operation. The container **120** is driven, for example, by a chain drive, for which purpose the container **120** in

this example has a flange **134** with a crown gear **136**, into which a chain (not shown) can engage in order to rotate the container **20**. A bearing flange **132** is arranged on the container **120** at an axial distance therefrom.

(37) The container **120** is mounted and guided in the framework **200** by the bearing flange **132**. For example, the container **120** has the bearing flange **132** at one axial end and the flange **134** with the crown gear **136** at the opposite axial end.

(38) The magnetic tape **52** can be arranged on the bearing flange **132** of the container **120** with which the container **120** is supported in the framework **200**.

(39) The magnetic tape **54** can be fastened to the bearing flange **132** on an outer side of a suitably bent holding sheet plate **60**, for example made of aluminum, and enclose the container **120** on its outer side. The holding sheet plate **60** can be arranged, for example, in a free space on the inside of the bearing flange **132**.

(40) The holding sheet plate **60** can consist of or comprise individual parts that are bent on a bending machine. The holding sheet plate **60** can then be fastened to the inside of the bearing flange **132**, for example by gluing. Then the magnetic tape **52** can be fastened to the holding plate **60**, for example by gluing, or fixed with a tightening strap or in another suitable manner.

(41) The sensor **54** is mounted at a small distance from the magnetic tape **52**. An advantageous distance is in particular between 1 mm and 3 mm. For this purpose, the sensor **54** can be arranged on a holding arm **56** which points towards the magnetic tape **52** and is fixed to the framework **200** (FIG. 6). In this way, the sensor **54** can be positioned in a stable and reproducible manner in relation to the magnetic tape **52**.

(42) The connections of holding arm **56** and sensor **54** to the framework **200** can advantageously be detachable and can be designed, for example, as screw connections.

(43) FIG. 9 shows a detailed view of a magnetic tape **52** of a position transducer **50** on a drive flange **134** of a container **120** according to an exemplary embodiment of the invention. FIG. 10 shows a detailed view of the position transducer **50** on the drive flange **134** of the container **120** according to FIG. 9.

(44) The drive flange **134** is part of the container **120** itself. In this embodiment, the magnetic tape **52** is attached to the outside of the drive flange **134**, for example by gluing.

(45) The sensor **54** is mounted at a small distance from the magnetic tape **52**. An advantageous distance is in particular between 1 mm and 3 mm. For this purpose, the sensor **54** can be arranged on a holding arm **56** which points towards the magnetic tape **52** and which is fixed to the framework **200** (FIG. 10). In this way, the sensor **54** can be positioned in relation to the magnetic tape **52** in a stable and reproducible manner.

(46) The connections of the holding arm **56** and the sensor **54** to the frame **200** can advantageously be detachable and can be designed, for example, as screw connections.

(47) It goes without saying that the magnetic tape **52** can also be arranged at a different location on the container **120**.

(48) FIG. 11 shows a cut-out perspective view of a container **120** in a framework **200** with an optical device **30** of a distance measuring device **10** on a holder **20** according to an exemplary embodiment of the invention.

(49) The holder **20** protrudes into the interior of the container **120** with an arm **22** pointing in the axial direction of the container **120**. The optical device **30** is fastened to the arm **22**, for example by screwing, tying, clamping or the like.

(50) The arm **22** is attached to a crossbeam **24** located in front of the aperture **126** of the container **120** and attached to the framework **200**. Typically, during the operation of the container **120**, solar radiation is directed into the container **120** through the aperture **126**.

(51) In particular, the crossbeam **24** can extend from one side of the framework **200** to the other side and cover the aperture **126**.

(52) The optical device **30** can be arranged to be displaceable along the arm **22**. Alternatively or

additionally, the arm **22** can be arranged to be displaceable on the crossbeam **24**.

(53) Alternatively or additionally, the arm **22** can also be designed as a telescopic arm.

(54) All connections of the arm **22** are expediently designed as detachable connections, so that the aperture **126** can be freed again after the measurements.

(55) FIG. **12** shows a flow chart for carrying out a method according to an exemplary embodiment of the invention.

(56) The method for determining a distribution of heat transfer medium **150** on an inner wall **130** of a container **120** starts with step **S100**.

(57) In a first sequence **S200**, a first measurement takes place, in which the inner wall **130** of the container **120** is measured. In step **S202**, the optical device **30** of the distance measuring device **10** measures the distances between the optical device **30** and points on the measurement line **40** on the inner wall **130**, point by point along a measurement line **40**, while the position transducer **50** measures the rotational position of the container **120** (step **S204**). A first three-dimensional point cloud is generated in step **S206** by linking distance and rotational position. The respective distance in relation to the respective measurement line **40** results in a height profile of the inner wall **130** both in the direction of the longitudinal axis **160** of the container **120** and, since the container **120** rotates, in the circumferential direction **140**.

(58) A second measurement takes place in a second sequence **S300**, in which the film **152** of the heat transfer medium **150**, which is distributed on the inner wall **130**, is measured. In step **S302**, the optical device **30** of the distance measuring device **10** measures the distances between the optical device **30** and points on the measurement line **40** on the film **152**, point by point along a measurement line **40**, while the position transducer **50** measures the rotational position of the container **120** (step **S304**).

(59) A further three-dimensional point cloud is generated in step **S306** by linking distance and rotational position. The respective distance in relation to the respective measurement line **40** results in a height profile of the film **152** on the inner wall **130** both in the direction of the longitudinal axis **160** of the container **120** and, since the container **120** rotates, in the circumferential direction **140**.

(60) It goes without saying that the measurement on the inner wall **130** can also be carried out after the measurement on the film **152**, so that the two sequences **S200** and **S300** can be swapped.

(61) In step **S402**, in sequence **S400**, a difference between the distance from the measurement on the inner wall **130** of the container **120** without heat transfer medium **150** and the distance from the measurement on the film **152** on the inner wall **130** is formed for each measuring point. In step **S404** a corrected point cloud is provided.

(62) After sequence **S400**, in step **S500**, measured values for the thickness of the film **152** corresponding to the distribution of the heat transfer medium **150** on the inner wall **130** are provided at a defined number of measuring points. The thickness of the film **152** or the distribution of the heat transfer medium **150** on the inner wall **130** can be determined with high spatial resolution.

(63) The method ends in step **S600**.

(64) The distance measurements are made without solar radiation entering the container **120**. The measurements are expediently carried out at comparable temperatures.

(65) Furthermore, corrections can be made during a reference measurement and during the measurement on the film **152** if the container **120** has deviations from circularity and/or an eccentricity. In particular, the measurement lines **40** can be adapted.

(66) The determination of the distribution of heat transfer medium **150** on the inner wall **130** can be carried out repeatedly at time intervals for quality control and any changes in the distribution can be detected. If permissible tolerances of the changes are exceeded, a maintenance requirement can be indicated.

LIST OF REFERENCE NUMERALS

(67) 10 distance measuring device 20 holder 22 arm 24 crossbeam 30 optical device 32 transmitter 34 receiver 40 measurement line 42 region 50 position transducer 52 magnetic tape 54 sensor 56 arm 60 holding sheet plate 90 computing device 100 receiver device 112 solar radiation 114 angle 120 container 122 housing 124 medium inlet 126 aperture 130 inner wall 132 bearing flange 134 drive flange 136 crown gear 140 circumferential direction 150 heat transfer medium 152 film 154 height difference 160 longitudinal axis 162 axial extent 170 direction of rotation 200 framework

## Claims

1. A measuring device for determining a distribution of a heat transfer medium on an inner wall of a shaftless container which rotates when used as intended and which is designed in particular to heat the heat transfer medium with concentrated solar radiation in a solar thermal power plant or as a rotary kiln, comprising a distance measuring device for determining a thickness of a film of the heat transfer medium on the inner wall of the container, wherein the distance measuring device has at least one optical device for detecting at least one height profile along at least one measurement line projected onto the inner wall and at least one position transducer for determination of a current rotational position of the respective measurement line on the inner wall.
2. The measuring device according to claim 1, wherein the distance measuring device is designed to detect distance data directly on the inner wall of the container along the measurement line in order to create the height profile.
3. The measuring device according to claim 1, wherein the distance measuring device is designed to detect distance data directly on the film of the heat transfer medium on the inner wall of the container along the measurement line in order to create the height profile.
4. The measuring device according to claim 1, wherein the measurement line extends parallel to a longitudinal axis of the container.
5. The measuring device according to claim 1, wherein the position transducer is attached to the container and detects a rotational position of the container, in particular synchronously with the projected measurement line.
6. The measuring device according to claim 5, wherein the position transducer has a magnetic tape which is placed around the container, in particular wherein the rotational position is determined via a magnetic tape comprising a number of magnetic poles which are read out by a sensor of the position transducer.
7. The measuring device according to claim 1, wherein a computing device is coupled to the distance measuring device and/or the position transducer.
8. The measuring device according to claim 1, wherein the distance measuring device protrudes into the container on a holder.
9. A method for determining a distribution of a heat transfer medium on an inner wall of a shaftless container that rotates when used as intended, in particular in which the heat transfer medium is heated with concentrated solar radiation in a solar thermal power plant or a rotary kiln, by means of a measuring device according to claim 1, wherein the distance measuring device detects at least one height profile along at least one measurement line projected onto the inner wall by means of at least one optical device and at least one position transducer determines a current position of the respective measurement line on the inner wall.
10. The method according to claim 9, wherein a position of the two height profiles with and without a heat transfer medium on the inner wall of the container is determined in relation to a rotational position of the container.
11. The method according to claim 9, wherein the distance measurement is carried out without solar radiation entering the container.
12. The method according to claim 9, wherein a difference between distance data of the film and distance data of the inner wall is formed and from this difference a position-dependent distribution

of the film of the heat transfer medium on the inner wall is determined.

13. The method according to claim 9, wherein a measuring frequency of the position transducer is adapted to a length of the measurement line in the axial direction of the container.
  14. The method according to claim 9, wherein a reference measurement is carried out to determine an eccentricity and/or lack of circularity of the container and a length of the measurement lines in the direction of the longitudinal axis of the container is adapted.
  15. The method according to claim 9, wherein the determination of the distribution of heat transfer medium on the inner wall is carried out repeatedly and changes in the distribution are detected, in particular wherein a maintenance requirement display is indicated if permissible tolerances of the changes are exceeded.
  16. Use of a measuring device for determination of a distribution of a heat transfer medium on an inner wall of a container which rotates when used to heat the heat transfer medium with concentrated solar radiation in a solar thermal power plant or in a rotary kiln, the measuring device comprising a distance measuring device for determination of a thickness of a film of the heat transfer medium on the inner wall of the container, detecting, using at least one optical device, at least one height profile along at least one measurement line projected onto the inner wall and determining, using at least one position transducer, a current rotational position of the respective measurement line on the inner wall, wherein the position transducer has a combination of a magnetic tape and a sensor for detecting a rotational position of the shaftless rotating container.
  17. Use according to claim 16, wherein the magnetic tape is arranged on the container and rotates with the container, and the sensor is arranged above the magnetic tape.
  18. Use according to claim 16, wherein the sensor is arranged on the container and rotates with the container and the magnetic tape is arranged above the sensor.
  19. Use of a combination of a magnetic tape and a sensor arranged above it to detect a rotational position of a shaftless rotating container in a solar thermal power plant or a rotary kiln.
  20. Use according to claim 19, wherein the magnetic tape is arranged on the container and rotates with the container and the sensor is arranged above the magnetic tape.
  21. The method according to claim 19, wherein the sensor is arranged on the container and rotates with the container and the magnetic tape is arranged above the sensor.
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